

Lab

Java basics — getting started

Lab: Introduction to Java and Eclipse

Objective

In this lab, you will:

- Create a Console program using the Eclipse IDE.
- Run the program.

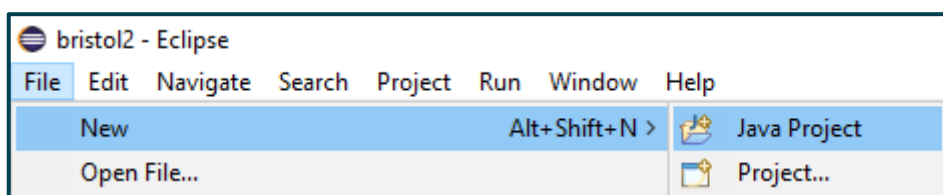
Overview

Eclipse is the development tool used by many Java developers and throughout this course. This lab will show you how to create and run a Java program.

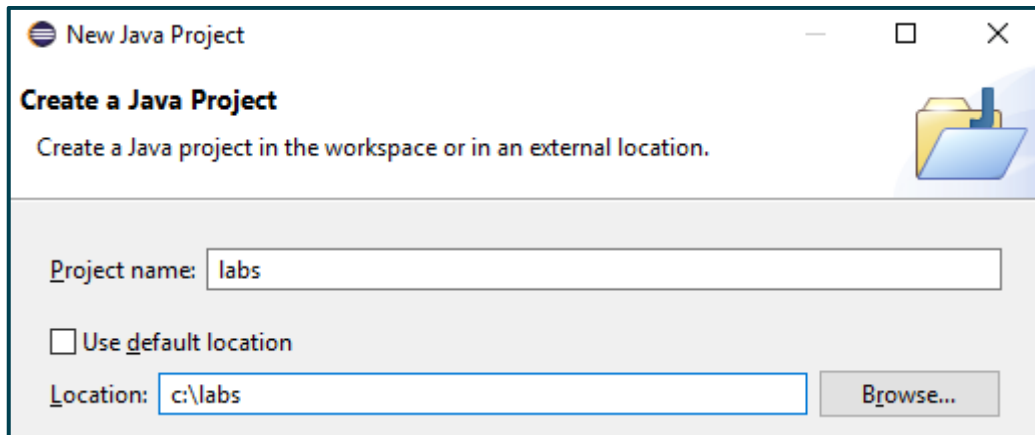
Step-by-step

Creating a new Java application

1. Launch Eclipse and type C:\Labs as the location for your 'workspace'. (You may choose a different location but make a note of where it is.)
2. Dismiss the Welcome window if it appears.
3. Choose menu option File > New > Java Project.



4. Type the following project name. If the default location is the one you want to use, leave it as it is; otherwise, uncheck 'Use default location' and enter your choice of folder. Accept the other default settings:



5. Select 'Finish'.
6. Right Select on the **labs** > **src** node and select **New** > **class**.
7. Type **lab1** as *package* name and select **public static void main(String[] args)**.



Select 'Finish'. (This step will be left out from now on.)

Writing the code

8. In the Program class's main method, type the following:

```
package lab1;

public class Program {
    public static void main(String[] args) {
        System.out.println("Hello, world!");
    }
}
```

Tip: Type **syso** followed by Ctrl-Space to type **System.out.println**

Make sure you make full use of this time saving editor command.

Building and running the program

9. Compiling the program is done by saving your code (Ctrl-s).
10. Press **Ctrl+F11** to run the program or select the green Run icon.
A Console window will become active at the bottom of Eclipse.
Please view the output for the **println** messages.

Building and running the program using command line

In this part you will deploy a simple Java app but the procedure is the same for a larger apps through to production: **Develop → Compile → Stage → Deploy**

Let's define a few terms:

- **Development:** Where you write and test code on your own machine
- **Staging:** A "dress rehearsal" environment that mimics production.
You deploy here first.
- **Production:** The live environment users interact with.

Use notepad to write a simple app (file: HelloWorld.java)

```
public class HelloWorld {  
    public static void main(String[] args) {  
        System.out.println("Hello, World!");  
    }  
}
```

Use command line to:

Compile: `javac HelloWorld.java`

Package into a JAR: `java -cfe HelloWorld.jar HelloWorld HelloWorld.class`

Run the code: `java -jar HelloWorld.jar`

Store your files in Github

Objectives:

In this section, you will store all your files using Git commands and use the same method to push your files to a cloud repository on GitHub.

1. Install Git on your PC if you've not installed it already.

Please select the links below for Mac or windows PCs.

(Alternatively, use QA's provided Virtual PCs.)

<https://git-scm.com/download/mac>

<https://git-scm.com/download/win>

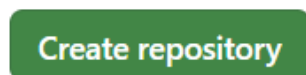
2. Create a free GitHub account using <https://github.com/> if you don't have one.



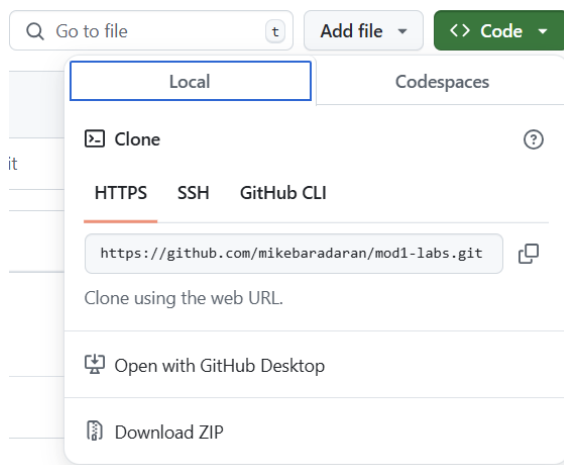
3. Create a public new repository called **mod1-labs** (or any name you prefer).



4. Select **Create repository**.



5. Copy the address of your repository.



6. Start a new Terminal session and navigate to the **\labs** folder.

```
cd \labs
git add .
git commit -m "first commit"
git branch -M Main
git remote add origin https://github.com/YourUsername/YourRepo.git
git push -u origin Main
```

7. To save all the files in the **\labs** folder each time you complete a lab, type these commands.

Open a Terminal session in \labs folder

```
git add .
git commit -m "lab xx completed"
git push
```

8. Now, refresh your browser to view your lab files on GitHub.



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